

GOSFORD HIGH SCHOOL



MATHEMATICS

Higher School Certificate

2009

Half Yearly Examination

Time Allowed: 2 Hours + 5 minutes reading time

Instructions:

- Start each **SECTION** in a new answer booklet, with each question on a new page.
- Write on one side of the page only.
- Write your name and number on each booklet.
- All questions are of equal value.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board approved calculators may be used in all sections.
- A table of standard integrals is provided.

SECTION 1 (Start a new booklet)

Question 1: (8 Marks)
Marks

a. Evaluate, correct to 3 significant figures:

i. $4e^{-1.5}$ 1

ii. $\log_e 30$ 1

b. Find a primitive of $5e^{4x}$ 1

c. Factorise fully:
 $xe^{2x} - 5xe^x$ 1

d. If $y = \ln(3x + 2)$, find $\frac{dy}{dx}$ 1

e. If $x^{4.7} = 4$, evaluate $\log_x 16$ 1

f. Sketch, on a quarter page diagram, the graph of $y = 3^{-x}$ 2
 (Show essential features with **TWO** points marked on the curve.)

SECTION 1 (Continued)

Question 2: (8 Marks) (Start a new page)

Marks

a. Find:

i. $\int x^5 - 2x^3 - 4 \, dx$

1

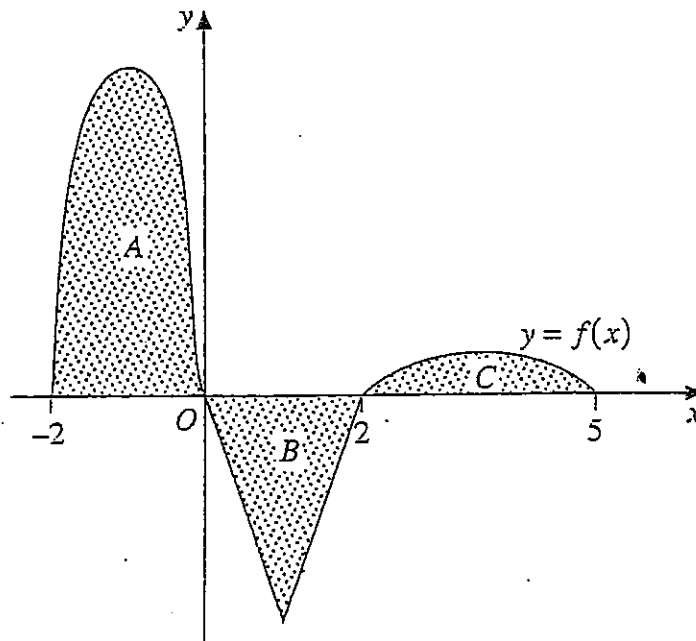
ii. $\int \frac{4}{x^3} \, dx$

1

iii. $\int_0^1 (3x+2)^4 \, dx$

2

b.



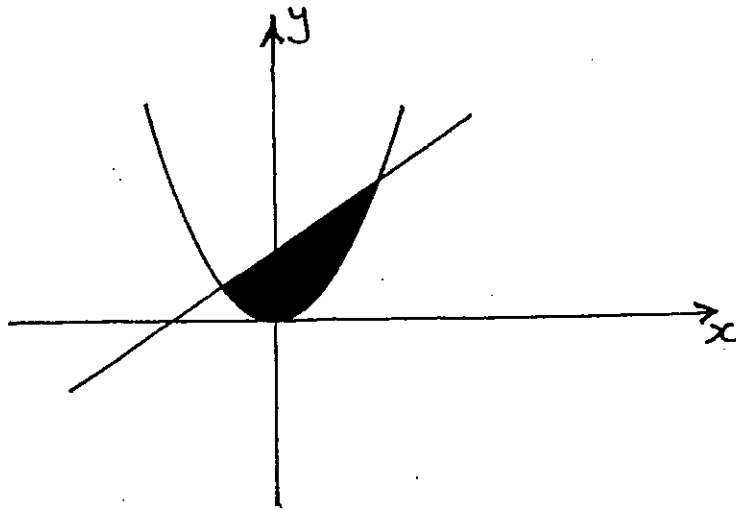
The graph of the function f is shown in the diagram. The shaded areas are bounded by $y = f(x)$ and the x axis. Shaded area A is 8 square units, shaded area B is 5 square units and shaded area C is 2 square units.

Evaluate $\int_{-2}^5 f(x) \, dx$

1

SECTION 1 (*Continued*)**Question 2:** (*Continued*)

- c. The graphs $y = x^2$ and $y = x + 2$ are shown below:



- i. Find the co-ordinates of the points of intersection of the two graphs. 1
- ii. Hence find the area of the shaded region. 2

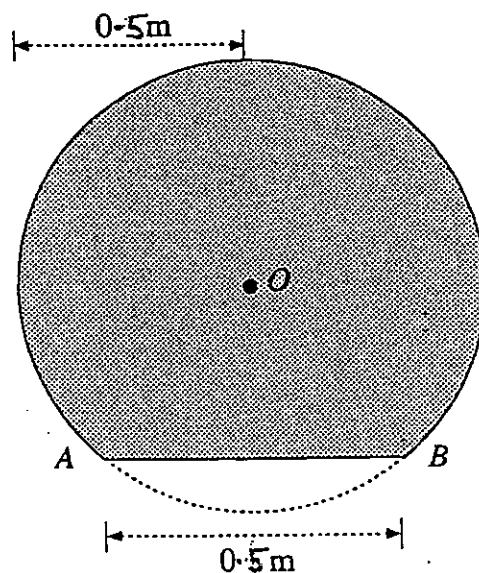
SECTION 2 (Start a new booklet)**Question 3:** (8 Marks)

	Marks
a. If $y = e^{\frac{x}{3}}$ and $\frac{dy}{dx} = ky$, find k .	1
b. Differentiate with respect to x :	
i. xe^x	1
ii. $\frac{x}{\ln x}$	2
iii. $(\log_e x)^3$	1
c. Find:	
i. $\int \frac{2x}{x^2+1} dx$	1
ii. $\int_1^e \frac{1}{2x} dx$, answer in simplest exact form	2

SECTION 2 (Continued)

Question 4: (8 Marks) (Start a new page)

- | | Marks |
|---|--------------|
| a. Express 105° in radians, giving your answer in simplest exact form. | 1 |
| b. Change 1.25 radians into degrees and minutes, to the nearest minute. | 1 |
| c. A sector is formed from a circle of radius 5 cm by subtending an angle of $\frac{\pi}{6}$ radians at the centre. Find, in exact form, the: | |
| i. Arc length of the sector | 1 |
| ii. Area of the sector | 1 |
| d. | |



A table-top is in the shape of a circle with a small segment removed as shown. The circle has centre O and radius 0.5 metres. The length of the straight edge AB is also 0.5 metres.

- | | |
|--|---|
| i. Explain why $\angle AOB = \frac{\pi}{3}$ radians. | 1 |
| ii. Find the area of the table top, giving your answer to 3 decimal places | 3 |

SECTION 3 (Start a new booklet)**Question 5:** (8 Marks)**Marks**

a. If $f(x) = x^4 - 2x^3 + 12$, find the value of:

i. $f(-1)$

1

ii. $f'(-1)$

1

iii. $f''(-1)$

1

b. The point (3,-11) is a turning point on the curve $y = ax^2 + bx + 7$.
Find the values of a and b .

3

c. Show that $y = \frac{1}{(2x-1)^3}$ is monotonic decreasing for all x , except $x = \frac{1}{2}$

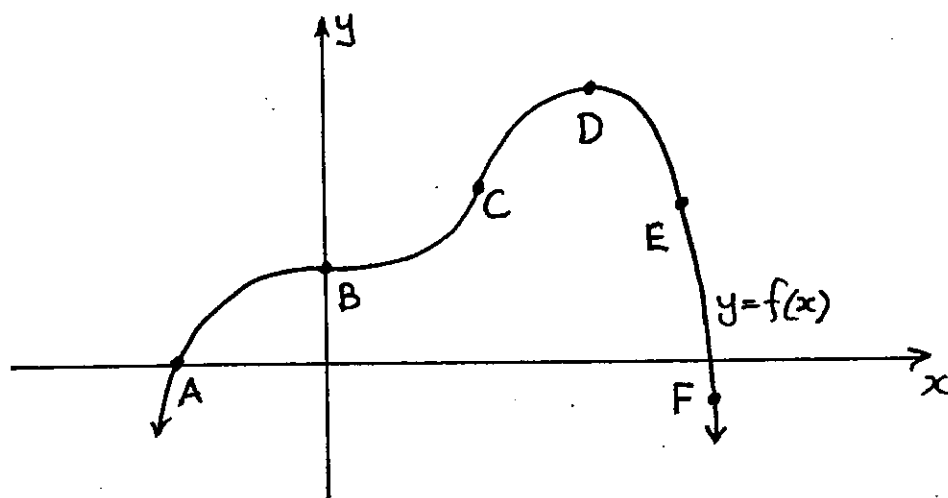
2

SECTION 3 (Continued)

Question 6: (8 Marks) (Start a new page)

Marks

- a. The diagram below shows the graph of $y = f(x)$, with 6 points on it A, B, C, D, E and F marked.



Write down the letter(s) which correspond to the point(s) where:

- | | | |
|------|--|---|
| i. | $f(x) = 0$ | 1 |
| ii. | $f'(x) = 0$ | 1 |
| iii. | $f''(x) = 0$ | 1 |
| iv. | $f'(x) < 0$ | 1 |
| b. | Copy the graph in part (a) into your answer booklet and draw the graph of $y = f'(x)$ on the same axes. (Clearly label your graphs.) | 2 |
| c. | The gradient function of a curve is given by $\frac{dy}{dx} = 5x - 6$.
If the curve passes through $(1, -3)$ find the equation of the curve. | 2 |

SECTION 4 (Start a new booklet)

Question 7: (8 Marks)
Marks

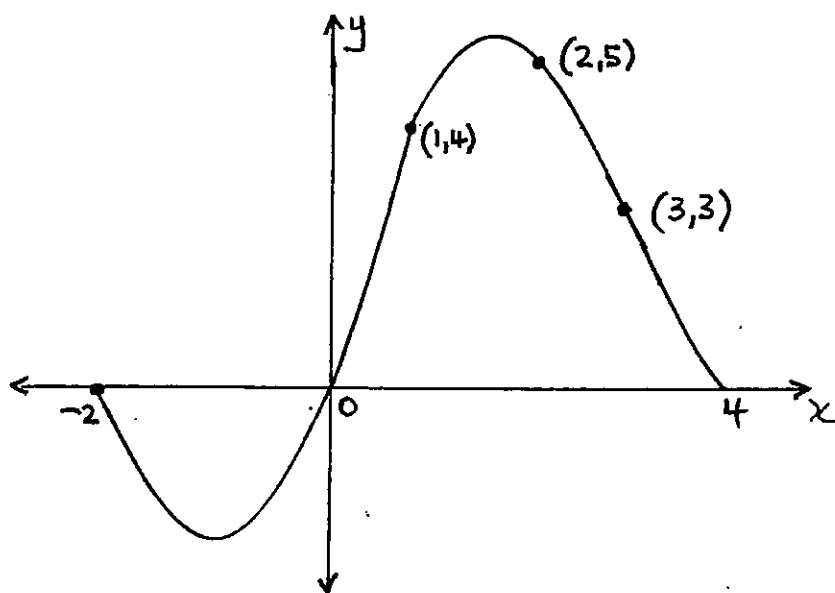
- a. Find the exact volume generated when the curve $y = \sqrt{x+3}$, between $x = -2$ and $x = 2$ is rotated about the x axis.

2

- b. Find the area enclosed between the curve $y = x^3$, the y axis and the lines $y = 1$ and $y = 8$.

2

- c. The diagram below is the graph of $y = f(x)$ for $-2 \leq x \leq 4$

(Not to scale)


- i. Write down an expression for the exact area bounded by the curve $y = f(x)$ and the x axis. (You are **not** required to find the equation of the curve.)
- ii. Use Simpson's Rule with 5 function values to approximate the area enclosed by the curve, the x axis and the lines $x = 0$ and $x = 4$

1

3

SECTION 4 (Continued)

Question 8: (8 Marks) (Start a new page)**Marks**

- a. i. Complete the table of values for $y = 2^x$

x	-1	0	1	2
y				

1

- ii. Using the Trapezoidal rule and the table in Part (i), approximate the area enclosed by the curve $y = 2^x$, the x axis and the lines $x = -1$ and $x = 2$

2

- b. Show that a primitive of $x\sqrt{x}$ is $\frac{2x^2\sqrt{x}}{5}$

2

- c. Find the area between the curve $y = x^2 - 4$, the x axis and the ordinates $x = 1$ and $x = 3$

3

SECTION 5 (Start a new booklet)

Question 9: (8 Marks)**Marks**

- a. Prove that the equation of the normal to the curve $y = \ln\left(\frac{1}{2x}\right)$ at the point where $x = e$ is given by:

3

$$y = ex - e^2 - \ln 2 - 1$$

- b. Given $\log_8 81 - \log_2(\sqrt[3]{3}) = \log_x y$
find the values of x and y .

2

- c. Find the stationary points on the curve $y = x^3 e^x$
and determine their nature.

3

SECTION 5 (Continued)**Question 10:** (8 Marks) (Start a new page)**Marks**

- a. A farmer is building a wheat silo in the shape of a closed cylinder of radius r metres and height h metres. The silo is to be made from galvanised iron sheeting and is to have a capacity of 300m^3 .

Using the formulae $V = \pi r^2 h$ and
 $S = 2\pi r^2 + 2\pi rh$

- i. Find an expression for h in terms of r .

1

- ii. show that the surface area S , is given by $S = \frac{2\pi r^3 + 600}{r}$

1

- iii. Hence, find the value of r , in exact form, that gives a minimum area of galvanised iron sheeting to be used.

2

- b. For a continuous curve $y = f(x)$ it is known that:

- The curve has only one x intercept
- $\frac{d^2y}{dx^2} = 36x^2 - 96x + 48$
- $(0,0)$ and $(2,16)$ are stationary points

- i. Find all x values that have points of inflexion.

3

- ii. Sketch the curve $y = f(x)$, showing all intercepts and stationary points.

1

END OF TEST

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Q1 a. (i) $4e^{-1.5}$
 $= 0.89252... \text{ (calc)}$
 $= 0.893 \text{ (3.s.f)}$

(ii) $\log_e 30$
 $= 3.401197... \text{ (calc)}$
 $= 3.40 \text{ (3.s.f)}$

b. Primitive $= \frac{5}{4} e^{4x}$

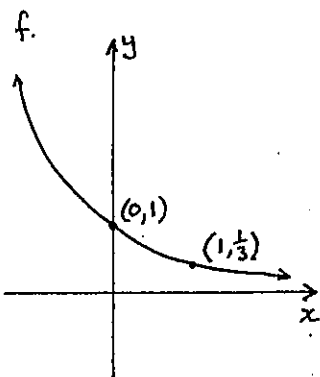
c. $xe^{2x} - 5xe^x$
 $= xe^x(e^x - 5)$

d. $y = \ln(3x+2)$
 $\frac{dy}{dx} = \frac{3}{3x+2}$

e. $x^{4.7} = 4$

$\therefore \log_x 4 = 4.7$

$\therefore \log_x 16 = 2 \log_x 4$
 $= 9.4$



Q2 a(i) $\int x^5 - 2x^3 - 4x dx$
 $= \frac{x^6}{6} - \frac{2x^4}{4} - 4x + C$

(ii) $\int 4x^{-3} dx$
 $= -\frac{4}{2} x^{-2} + C$
 $= -\frac{2}{x^2} + C$

(iii) $\int_0^1 (3x+2)^4 dx$
 $= \frac{1}{5} \left[\frac{(3x+2)^5}{3} \right]_0^1$
 $= \frac{1}{15} (5^5 - 2^5)$
 $= \frac{3093}{15} = \frac{1031}{5}$

b. $\int_{-2}^5 f(x) dx$
 $= 8 + (-5) + 2$
 $= 5$

c. (i) $y = x^2$ - (1)
 $y = x+2$ - (2)

Solving simultaneously

Sub (1) in (2): $x^2 = x+2$
 $\therefore x^2 - x - 2 = 0$
 $(x-2)(x+1) = 0$
 $\therefore x = 2, -1$
 $y = 4, 1$

$\therefore$ Pts of intersection are (2, 4) and (-1, 1)

(ii) $A = \int_{-1}^2 x+2-x^2 dx$
 $= \left[\frac{x^2}{2} + 2x - \frac{x^3}{3} \right]_{-1}^2$
 $= (2+4-\frac{8}{3}) - (-\frac{1}{2}+2-\frac{1}{3})$
 $= 4\frac{1}{2} \text{ units}^2$

Q3 a. $y = e^{x/3}$
 $\frac{dy}{dx} = \frac{1}{3} e^{x/3}$
 $= \frac{1}{3} y$
 $\therefore k = \frac{1}{3}$

b. (i) let $y = xe^x$
 $y' = e^x \cdot 1 + x \cdot e^x$
 $= e^x(1+x)$

(ii) $y = \frac{x}{\ln x}$
 $y' = \frac{\ln x \cdot 1 - x \cdot \frac{1}{x}}{(\ln x)^2}$
 $= \frac{\ln x - 1}{(\ln x)^2}$

(iii) $y = (\log_e x)^3$
 $y' = 3(\ln x)^2 \cdot \frac{1}{x}$
 $= \frac{3(\ln x)^2}{x}$

c. (i) $\int \frac{2x}{x^2+1} dx$
 $= \ln(x^2+1) + C$

(ii) $\int_1^e \frac{1}{2x} dx$
 $= \frac{1}{2} [\ln x]_1^e$
 $= \frac{1}{2} [\ln e - \ln 1]$
 $= \frac{1}{2} (1 - 0)$
 $= \frac{1}{2}$

Q4 a. $105^\circ = \frac{105\pi}{180}$
 $= \frac{7\pi}{12} \text{ radians}$

b. $1.25 \times \frac{180}{\pi}$
 $= 71.6197... \text{ (calc)}$
 $= 71^\circ 37'$

c. (i) $l = r\theta$
 $= 5 \times \frac{\pi}{6}$
 $= \frac{5\pi}{6} \text{ cm}$

(ii) $A = \frac{1}{2} r^2 \theta$
 $= \frac{1}{2} \times 25 \times \frac{\pi}{6}$
 $= \frac{25\pi}{12} \text{ cm}^2$

d. (i) ΔAOB is equilateral
 $\therefore \angle AOB = 60^\circ$
 $= \frac{\pi}{3} \text{ radians}$
 as required

(ii) Area of table top
 $= \text{Area of Circle} - \text{Area of minor segment}$
 $= \pi r^2 - \frac{1}{2} r^2 (\theta - \sin \theta)$
 $= 0.25\pi - 0.125 \left(\frac{\pi}{3} - \sin \frac{\pi}{3} \right)$
 $= \frac{\pi}{4} - \frac{1}{8} \left(\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right)$
 $= \frac{\pi}{4} - \frac{\pi}{24} + \frac{\sqrt{3}}{16}$
 $= 0.7627516... \text{ (calc)}$
 $= 0.763 \text{ m}^2 \text{ (3d.p)}$

Q5 a. $f(x) = x^4 - 2x^3 + 12$
 (i) $f(-1) = (-1)^4 - 2(-1)^3 + 12$
 $= 1 + 2 + 12$
 $= 15$
 (ii) $f'(x) = 4x^3 - 6x^2$
 $f'(-1) = 4(-1)^3 - 6(-1)^2$
 $= -4 - 6$
 $= -10$

(iii) $f''(x) = 12x^2 - 12x$
 $f''(-1) = 12(-1)^2 - 12(-1)$
 $= 12 + 12$
 $= 24$

b. $y = ax^2 + bx + 7$
 T.P's occur when $y' = 0$
 $y' = 2ax + b$
 $\therefore 2ax + b = 0$

Since $(3, -11)$ is a T.P
 $6a + b = 0$ - (1)

Also $(3, -11)$ is a point on the curve
 $\therefore -11 = 9a + 3b + 7$
 $\therefore 9a + 3b = -18$ - (2)

Solving simultaneously

(1) $\times 3$: $18a + 3b = 0$ - (3)

(2) - (3) $-9a = -18$
 $\therefore a = 2$

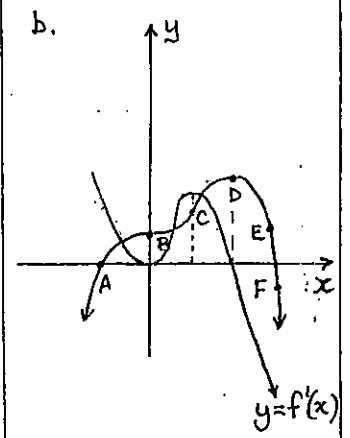
$b = -12$

c. If $y = \frac{1}{(2x-1)^3}$
 $x \neq \frac{1}{2}$
 $y = (2x-1)^{-3}$
 $y' = -3(2x-1)^{-4} \times 2$
 $= \frac{-6}{(2x-1)^4}$

Since $(2x-1)^4 > 0$ for all x except $x = \frac{1}{2}$
 $\therefore y' < 0$ for all x except $x = \frac{1}{2}$
 $\therefore$ always monotonic decreasing.

Q6 a:

- (i) A
- (ii) B and D
- (iii) C and B
- (iv) E and F



c. $\frac{dy}{dx} = 5x - 6$
 $y = \frac{5x^2}{2} - 6x + C$
 when $x=1, y=-3$
 $-3 = \frac{5}{2} - 6 + C$
 $\frac{1}{2} = C$

$\therefore$ equation of curve is $y = \frac{5x^2}{2} - 6x + \frac{1}{2}$

Q7 a. $V = \pi \int y^2 dx$
 $V = \pi \int_{-2}^2 x + 3 dx$

$= \pi \left[\frac{x^2}{2} + 3x \right]_{-2}^2$
 $= \pi \left[(2+6) - (-2-6) \right]$
 $= \pi (8+4)$
 $= 12\pi \text{ units}^3$
 b. $A = \int_1^8 y^{\frac{1}{3}} dy$
 $= \frac{3}{4} \left[y^{\frac{4}{3}} \right]_1^8$
 $= \frac{3}{4} (16-1)$
 $= \frac{45}{4} \text{ units}^2$

c. (i) $A = \left| \int_{-2}^0 f(x) dx \right| + \left| \int_0^4 f(x) dx \right|$

x	0	1	2	3	4
y	0	4	5	3	0
y_0	y_1	y_2	y_3	y_4	

$A \doteq \frac{h}{3} [y_0 + y_4 + 4(y_1 + y_3) + 2y_2]$
 $= \frac{1}{3} [0 + 0 + 4(4+3) + 10]$
 $= \frac{38}{3} \text{ units}^2$

Q8 a.

(i)

x	-1	0	1	2
y	$\frac{1}{2}$	1	2	4
y_0	y_1	y_2	y_3	

 (ii) $A \doteq \frac{h}{2} [y_0 + y_3 + 2(y_1 + y_2)]$
 $= \frac{1}{2} \left[\frac{1}{2} + 4 + 2(1+2) \right]$
 $= \frac{21}{4} \text{ units}^2$

b. $\int x\sqrt{x} dx$
 $= \int x^{\frac{3}{2}} dx$
 $= \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + C$
 $= \frac{2}{5} \sqrt{x^5}$
 $= \frac{2x^2\sqrt{x}}{5} \text{ as req.}$

c.
 $A = \left| \int_1^2 (x^2 - 4x) dx \right| + \left| \int_2^3 (x^2 - 4x) dx \right|$
 $= \left| \left[\frac{x^3}{3} - 4x \right]_1^2 \right| + \left| \left[\frac{x^3}{3} - 4x \right]_2^3 \right|$
 $= \left| \left(\frac{8}{3} - 8 \right) - \left(\frac{1}{3} - 4 \right) \right| + \left| \left(9 - 12 \right) - \left(\frac{8}{3} - 8 \right) \right|$
 $= \left| -\frac{5}{3} \right| + \left| \frac{7}{3} \right|$
 $= 4 \text{ units}^2$

Q9 a. $y = \ln\left(\frac{1}{2x}\right)$
 $x=e$
 $y = -\ln 2e$
 $= -\ln 2 - 1$
 $= \ln(2e)^{-1}$
 $= -\ln 2e$
 $\therefore y' = -\frac{2}{2x}$
 $= -\frac{1}{x}$

when $x=e$
 $y' = -\frac{1}{e} = m_{\text{Tangent}}$ to determine nature
 $\therefore M_{\text{normal}} = e$
 equation of normal
 $\Rightarrow y - (-\ln 2 - 1) = e(x - e)$
 $y + \ln 2 + 1 = ex - e^2$
 $y = ex - e^2 - \ln 2 - 1$
 as required

b. $\log_8 81 - \log_2 \left(\frac{3}{\sqrt{3}}\right)$
 $= \frac{\log_2 81}{\log_2 8} - \log_2 3^{\frac{1}{2}}$
 $= \frac{\log_2 3^4}{\log_2 2^3} - \frac{1}{2} \log_2 3$
 $= \frac{4 \log_2 3}{3 \log_2 2} - \frac{1}{3} \log_2 3$
 $= \frac{4 \log_2 3}{3} - \frac{1}{3} \log_2 3$
 $= \log_2 3 \therefore x=2, y=3$

c. $y = x^3 e^x$
 Stat. pts. occur when $y' = 0$
 $y' = e^x \cdot 3x^2 + x^3 \cdot e^x$
 $= x^2 e^x (3+x)$
 $y' = 0$ when $x = 0, -3$
 $\therefore$ Stat. pts occur at $(0, 0)$ and $(-3, -\frac{27}{e^3})$
 at $(0, 0)$:

x	0	0	0	0
y'	+	+	0	+

 at $(-3, -\frac{27}{e^3})$:

x	-3	-3	-3	-3
y'	-	-	0	+

 $\therefore$ Min T.P at $(-3, -\frac{27}{e^3})$

Q10 a.
 (i) $V = \pi r^2 h$
 $300 = \pi r^2 h$
 $\frac{300}{\pi r^2} = h$
 $S = 2\pi r^2 + 2\pi r \left(\frac{300}{\pi r^2}\right)$
 $= 2\pi r^2 + \frac{600}{r}$
 $= \frac{2\pi r^3 + 600}{r}$
 as required

(iii) Min value when $S' = 0$
 $S' = 4\pi r - \frac{600}{r^2}$
 $= \frac{4\pi r^3 - 600}{r^2}$
 $S' = 0$ when $4\pi r^3 - 600 = 0$
 $4\pi r^3 = 600$
 $r^3 = \frac{150}{\pi}$
 $r = \sqrt[3]{\frac{150}{\pi}}$
 $S'' = 4\pi + \frac{1200}{r^3}$
 when $r = \sqrt[3]{\frac{150}{\pi}}$ $S'' > 0$
 $\therefore$ a minimum area when $r = \sqrt[3]{\frac{150}{\pi}}$
 b. Points of inflexion occur when $y'' = 0$
 $36x^2 - 96x + 48 = 0$
 $3x^2 - 8x + 4 = 0$
 $(3x-2)(x-2) = 0$
 $\therefore x = \frac{2}{3}, 2$
 Test for concavity change
 $x = \frac{2}{3}$

x	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$	$\frac{2}{3}$
y''	+	+	0	-

 $x = 2$

x	2	2	2	2
y''	-	-	0	+

 (H.I.) $\therefore (\frac{2}{3}, 0)$ is a pt. inflexion $(2, 16)$ is H.I.
 (ii)